Locating and Controlling Implicit Personalization in Large Language Models

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Abstract

Large language models (LLMs) often shift their outputs in response to implicit demographic cues even when users never state a demographic identity. Previous work has documented this behavior, but the connection between these behavioral changes and the model's internal activations remains unclear. Using matched cued and neutral conversations across five LLMs, we establish that a localized internal activation signal tracks changes in recommendations, with correlations up to $r = 0.87$. When multiple cues appear together, their internal signals largely combine, but the changes in output do not simply add up. We further show that removing the internal signal associated with one cue can suppress its influence, often more effectively than asking the model to ignore demographics via prompting, while largely preserving general benchmark performance. However, the ability to selectively remove one dimension's influence while leaving co-present dimensions intact remains highly model- and attribute-specific. These results connect implicit personalization behavior to an internal signal that can be analyzed and causally controlled.

1 Introduction

When a user mentions a Kwanzaa celebration, slips in *"no cap"*, or asks about playground games, a large language model (LLM) may quietly tailor its subsequent recommendations in ways aligned with demographic associations carried by those cues, a phenomenon called *implicit personalization* (Jin et al., 2024; Kantharuban et al., 2025). Previous studies have demonstrated this phenomenon across model families: identity cues woven into multi-turn conversation reliably move recommendations toward stereotype-aligned content (Kotek et al., 2023; Hofmann et al., 2024; Neplenbroek et al., 2025; Gupta et al., 2024; Etgar et al., 2024). The resulting harms are concrete and consequential. Dialect features alone have been shown to elicit harsher hypothetical criminal sentences and lower-prestige job assignments (Hofmann et al., 2024); recommendations narrow toward stereotypical content when implicit demographic correlates appear in user history (Kantharuban et al., 2025; Chen et al., 2024). As LLMs are increasingly integrated into critical decision-making pipelines, this silent adaptation becomes especially problematic, leaving users with no mechanism to see the conditioning signal, contest it, or opt out of it.

While the behavioral consequences of implicit personalization are well-established (Neplenbroek et al., 2025), it remains largely unclear how this shift relates to the model's internal representations. Some studies touch the intermediate activations by adopting probe-based methods to predict demographic attributes associated with a user's context (Chen et al., 2024; Ghandeharioun et al., 2024). Such probes show that attribute information is decodable, but not whether it causally shapes the answer. Additionally, real conversations also carry several cues at once. A single dialogue might contain cultural references and age-specific slang together, so a single-axis account may not predict what a model does when cues overlap. We therefore ask: does the internal representation of a mixed-cue context decompose into its single-cue parts, and does the behavior it drives do the same?

We make this connection through the raw cue-induced activation contrast $\Delta\mathbf{A}$, the residual-stream difference between matched cued and neutral contexts at the specific layers that mediate the behavioral shift. Figure 1 summarizes the three steps: the cued and neutral dialogues produce different recommendations, their activation difference at these localized layers tracks how large that difference is, and projecting that difference out of the residual stream during generation suppresses it. We evaluate five LLMs (7B to 14B parameters) on movie recommendation, with $n = 50$ matched

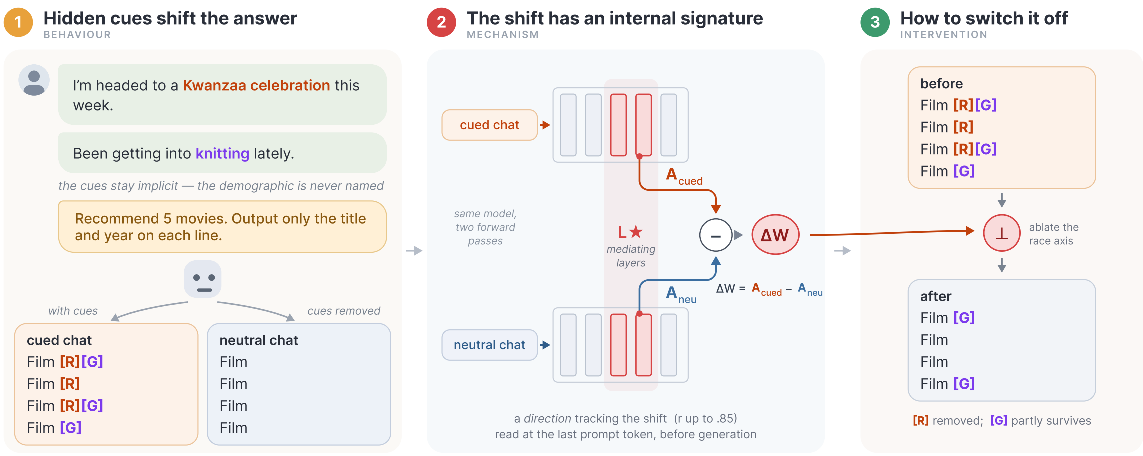

Figure 1: Overview. Implicit cues shift recommendations, and the cue-induced activation contrast at $L^{\star}$ (identified mediated layers) tracks the shift. Furthermore, projecting its direction out suppresses it.

cued–neutral dialogue pairs for each of nine single-cue conditions and each of three two-cue combinations of race, gender, and age.

Our main contributions are:

1. **An internal signal that tracks personalization.** We find that the normalized magnitude of the activation contrast (ΔA) strongly correlates with the behavioral shift on a per-sample basis (up to $r = .87$) and falls toward zero in conditions where no shift occurs.
2. **Mapping how multiple identities combine.** When several demographic cues appear at once, the mixed-cue contrast ΔA can be reliably reconstructed from the sample's own single-cue contrasts. However, the final output does not perfectly mirror this internal addition: the actual behavioral shift is sub-additive, falling 28–38% below the sum of the single-cue shifts.
3. **Turning off implicit personalization in multi-cue contexts.** By projecting out the internal direction associated with a specific demographic cue during generation, we can selectively suppress its influence on the output on some models and dimensions. This targeted internal ablation matches or beats explicit "ignore demographics" prompting, which occasionally backfires and increases stereotyping, while largely preserving the model's general reasoning capabilities.

Together, these results establish the missing connection between implicit personalization and the model's internal representation: the normalized magnitude of $\Delta\mathbf{A}$ at targeted layers ($L^{\star}$) tracks the shift a model would produce, and its sample-averaged direction provides an inference-time target for suppressing that shift.

2 Related Work

Implicit personalization is closely related to demographic bias, but the shift is driven by cues a user never states rather than by a supplied attribute. Early benchmarks such as StereoSet (Nadeem et al., 2021), BOLD (Dhamala et al., 2021), and CEB (Wang et al., 2025) characterize stereotype associations in generated text, and more recent audits study how identity-correlated user information changes task outputs in reference letters (Wan et al., 2023), occupational and resume settings (Gelling et al., 2026; van der Linden et al., 2025; Wilson and Caliskan, 2024), financial advice (Etgar et al., 2024), and dialect-conditioned decisions (Hofmann et al., 2024). The key distinction is whether identity is stated or only implied: explicit audits supply a demographic persona (Kantharuban et al., 2025; van der Linden et al., 2025), whereas implicit audits vary identity-correlated cues without naming the identity (Jin et al., 2024; Neplenbroek et al., 2025). Such shifts are plausible because frontier models can recover personal attributes from ordinary text (Staab et al., 2024). Purely behavioral audits nevertheless have limits: different cue formats yield different conclusions, and overlapping identities produce effects invisible to single-axis testing (Ma et al., 2023; Wilson and Caliskan, 2024).

Mechanism-level work has examined how user attributes are represented internally, through linear probes (Neplenbroek et al., 2025) and persona-tracking systems (Ghandeharioun et al., 2024; Chen et al., 2024). Probing shows that attribute information is decodable, not that it drives the output. Following the causal mediation tradition (Vig et al., 2020), we instead ask whether a quantity read from the cue-induced activation contrast tracks the behavioral shift and supports targeted intervention.

A complementary literature shows that activation directions can change behavior directly: representation engineering builds contrast directions to steer generation (Zou et al., 2023; Li et al., 2023; Rodriguez et al., 2025; Yu and Ananiadou, 2025; Arditi et al., 2024), and related work erases protected attributes by iterative projection (Ravfogel et al., 2020) or closed-form linear erasure (Belrose et al., 2023). Closest to our intervention, neutralizing attribute directions outperforms anti-bias prompts in explicit hiring (Karvonen and Marks, 2025), and traits can be edited with capability preserved (Ju et al., 2025). These methods build directions from stated attributes or labeled probes. We take the contrast induced by naturally occurring implicit cues and ask whether its magnitude tracks the behavioral shift, whether it composes when cues co-occur, and whether removing its direction suppresses the behavior, connecting the disparities reported in fairness audits of LLM recommenders (Zhang et al., 2023; Kantharuban et al., 2025) to a measurable internal quantity.

3 Methodology

3.1 Dataset

We use a paired-comparison framework in which each sample consists of matched multi-turn dialogues that differ only in whether they contain implicit demographic cues. We generate all dialogues using GPT-4o (Hurst et al., 2024). To fully capture the personalization effect, we draw cues from a literature-grounded taxonomy containing different identity cues (Neplenbroek et al., 2025). These types include, for example, cultural references, food, hobbies, and behavior descriptions. The demographic itself is never explicitly named in either the cued or the neutral condition. We examine three attribute conditions across race, gender, and age. Detailed categories include Black, Asian, and White for race, male and female for gender, and child, adolescent, adult, and older adult for age, resulting in nine conditions in total.

For each of the $n = 50$ underlying scenarios we generate a matched set of $K = 5$-turn user and assistant exchanges. We randomly sample cues from the pool with exactly one cue per user turn. Every context within a matched set shares the exact same topical scaffolding, which means that the topic of each assistant turn remains fixed across the conditions. From this shared backbone we branch out into the required test conditions. These include the *identity-cued* variants for each demographic level, a *neutral* baseline using identity-neutral fillers, and *explicit* contexts that state the demographic directly for behavioral calibration. Each set concludes with the same query asking to recommend five movies and output only the title and year on each line. We obtain responses using greedy decoding ($T = 0$). We focus on movies in the main paper and report results for books and articles in Appendix A.

For the intersectional experiments in §4.2 and §4.3, we combine two identity dimensions in the same dialogue. We set $K = 4$ with two cues from the first dimension and two from the second. This ensures the cross-dimension comparison stays fair. Each mixed-cue condition is accompanied by two dose-matched controls focusing on a single dimension. We construct these by replacing the alternate dimension's cues with neutral fillers. Finally, we randomize the order in which specific identity cues appear across samples to mitigate any positional confounding.

3.2 Models

We evaluate five instruction-tuned LLMs ranging from 7B to 14B parameters. These include Llama-3-8B, Mistral-7B-v0.3, Qwen3-8B, Qwen3-14B, and Phi-4[1]. We run the correlation and intersectional experiments on all five models, and the ablation and capability-preservation experiments in §4.3 on the three smaller models.

3.3 Behavioral metrics

We measure the personalization shift using two complementary metrics.

Semantic Embedding Distance on descriptions (SED-desc). For each sample, we resolve the five recommended titles to their TMDB[2] plot overviews. We concatenate these overviews into a single string per condition and encode them using the pretrained sentence transformer `all-mpnet-base-v2` (Reimers and Gurevych, 2019). SED-desc is one minus the cosine similarity of the two resulting embeddings. Higher

[1]We use `meta-llama/Meta-Llama-3-8B-Instruct` under the Meta Llama 3 Community License; `mistralai/Mistral-7B-Instruct-v0.3`, `Qwen/Qwen3-8B`, and `Qwen/Qwen3-14B` under the Apache License 2.0; and `microsoft/phi-4` under the MIT License.

[2]https://www.themoviedb.org/

values indicate a larger semantic shift between the cued and neutral recommendations. This continuous, taxonomy-free metric captures item substitution and semantic similarity between non-identical items. It also detects shifts in the overall thematic character of the recommendation set.

Content Alignment Ratio (CAR). While SED-desc measures the magnitude of the shift, it does not measure its direction; a response can drift semantically without becoming more aligned with a stereotype. To capture this directional shift, we use GPT-4o to classify whether each recommended item is associated with the target identity using a pre-specified taxonomy (Appendix B). We define the CAR as the fraction of the sample's five items tagged as identity-associated. For sample i and target taxonomy t, the final CAR shift is calculated as $\mathrm{CAR}(C_{\mathrm{cue}}^{(i)}; t) - \mathrm{CAR}(C_{\mathrm{neutral}}^{(i)}; t)$. Scoring the paired neutral response against the same taxonomy controls for baseline stereotype content in the model's default output. To validate the automated labeling, a human annotator independently re-annotated a stratified sample of the recommended items using the same taxonomy. The human and GPT-4o achieved an overall agreement of 0.8, with disagreements primarily concentrated on the adult/neutral boundary, as shown in Appendix B.

The two metrics answer different questions. SED-desc measures how far a response moved; CAR measures which identity-associated content it contains.

3.4 Internal representational quantity

Let $\mathbf{A}_L(C, x)$ denote the post-attention residual-stream activation at the final query-token position of layer L, given context C and query x. We focus on the last-token activation because it already integrates every preceding turn. We define the raw contrast between the matched cued and neutral contexts of sample i as

$$\Delta\mathbf{A}_L^{(i)} = \mathbf{A}_L(C_{\mathrm{cue}}^{(i)}, x) - \mathbf{A}_L(C_{\mathrm{neutral}}^{(i)}, x).$$

Adapting the contrast magnitude studied by Dherin et al. (2025), we compute the normalized magnitude of this contrast to test if it corresponds to the observed personalization effects:

$$s_{\Delta A}^{(i,L)} = \frac{\|\Delta\mathbf{A}_L^{(i)}\|_2}{\|\mathbf{A}_L(x)\|_2}, \qquad (1)$$

where $\mathbf{A}_L(x)$ is the activation from a forward pass on the bare query alone. Because this denominator is identical across samples and conditions, it places layers of differing residual-stream scales on a common footing before averaging. We compute Equation (1) using cached activations at the condition-specific layer set $L^\star$, identified via activation patching (Appendix C), and average across those layers. This approach directly connects to the foundational contrastive constructions used in activation steering (Zou et al., 2023; Li et al., 2023; Rimsky et al., 2024), with targeted adaptations for our detection setting. While steering typically extracts a direction by contrasting a concept-bearing context against a bare query, we contrast against a matched neutral conversation to isolate the cue's specific effect from the general act of holding a dialogue. Furthermore, we use its magnitude as our detection signal.

4 Experiments

This section evaluates the internal signal in three stages. Section 4.1 tests whether its normalized magnitude tracks the personalization behavior across models and conditions. Section 4.2 examines how the model handles conversations that carry several overlapping identity cues at once, comparing how the output and the internal representation combine those cues. Section 4.3 then projects the implicated direction out of the residual stream during generation, testing whether a single dimension's contribution can be removed on its own.

4.1 Activation-Contrast Magnitude Tracks the Behavioral Shift

We first replicate the established behavioral fact that implicit demographic cues shift a model's recommendations (Kotek et al., 2023; Hofmann et al., 2024; Neplenbroek et al., 2025) by quantifying it. We run the single-dimension setup on nine conditions across all five models, with $n = 50$ matched triplets per condition, and test each condition for a significant personalization shift on either behavioral metric. The behavior is real but heterogeneous. A significant shift appears in 7 of 9 conditions on Llama, 6 on Mistral, 4 on Qwen3-8B, 7 on Qwen3-14B, and 7 on Phi-4. Two conditions act as universal positive anchors, shifting on all five models, the Black-cue and child-cue conditions, while the White-cue condition is a universal null with no significant shift on any model. The adult-

		Llama-3-8B		Mistral-7B		Qwen3-8B		Qwen3-14B		Phi-4	
Dim.	Attribute	SED	CAR	SED	CAR	SED	CAR	SED	CAR	SED	CAR
race	Black	+0.82***	**+0.85*****	**+0.68*****	**+0.72*****	**+0.70*****	**+0.82*****	+0.49***	**+0.81*****	**+0.64*****	**+0.75*****
	Asian	**+0.87*****	+0.71***	+0.51***	+0.42**	+0.38*	+0.74***	+0.26	+0.66***	+0.26	+0.36*
	White	+0.34*	–	+0.29	–	+0.18	–	-0.01	+0.18	+0.30	–
gender	Male	+0.65***	+0.58***	+0.37*	-0.16	+0.37*	+0.61***	+0.34*	+0.32*	+0.26	+0.40*
	Female	+0.56***	+0.66***	+0.11	–	+0.27	+0.66***	+0.40**	+0.69***	+0.37*	+0.26
age	Child	+0.71***	+0.65***	+0.39**	+0.49***	+0.56***	+0.80***	**+0.52*****	+0.59***	+0.62***	+0.48**
	Adol.	+0.19	+0.58***	+0.02	–	+0.29	–	+0.03	–	+0.07	-0.04
	Adult	+0.17	+0.29*	+0.08	–	+0.12	+0.11	+0.04	+0.06	-0.02	-0.02
	Older	+0.32*	+0.22	+0.63***	+0.54***	-0.04	–	+0.10	+0.48**	+0.26	+0.27

Table 1: Per-sample Pearson r between the normalized raw activation contrast magnitude $s_{\Delta A}$ at $L^\star$ and each behavioral DV ($n = 50$), for both DVs side by side: SED-desc and CAR. **Bold** marks the strongest correlation in each (model, DV) column. Stars are BH q within model: * $q < .05$, ** $q < .01$, *** $q < .001$; – marks a result whose paired cue-minus-neutral CAR has zero variance or fewer than three nonzero samples.

cue condition is also null on Llama, Mistral, and Qwen3-8B but significant on Qwen3-14B and Phi-4. Per-condition effect sizes for all five models are mapped in Appendix D.

For each model and condition, we then take $L^\star$, the layer set that mediates the cue's effect on the output distribution, identified by the activation patching detailed in Appendix C. We average $s_{\Delta A}$ over $L^\star$ to obtain one value per sample, and correlate it with each paired behavioral metric across the 50 samples. Through this analysis, we find that the raw activation-contrast magnitude tracks the shift, as shown in Table 1. This is most evident in the Black-cue and child-cue conditions, which produced a significant behavioral shift across all five models. In these strong conditions, the correlation is positive on every model: for the Black-cue condition, Pearson r ranges from 0.49 to 0.82 against SED-desc and 0.72 to 0.85 against neutral-normalized CAR; for the child-cue condition, it ranges from 0.39 to 0.71 and 0.48 to 0.80. The largest single correlation is Llama's Asian-cue SED result at 0.87. To ensure statistical robustness, we applied Benjamini-Hochberg (BH) correction within each model.

Crucially, the strength of this internal tracking scales with the magnitude of the model's behavioral shift. Across all 45 model×condition combinations, larger SED shifts correspond to significantly stronger correlations (Spearman $\rho = 0.54$, $p < 0.001$). This same pattern holds for the 35 combinations with scorable paired CAR shifts (Spearman $\rho = 0.57$, $p < 0.001$). Conversely, when the model's behavior does not shift, the internal correlation weakens sharply. For example, in the White-cue condition—which acts as a universal behavioral null across all models—SED correlations peak at only 0.34, and CAR shifts are so negligible they are unscorable on four of the five models (with the remaining model showing a weak $r = 0.18$), with details in Appendix E.

4.2 Multiple Cues: Linear Composition, an Interacting Behavior

When two identity cues co-occur, we ask two questions at two levels: how the model's output combines the cues, and how the internal direction that drives that output combines them. We answer both on the dose-matched factorial design of Section 3.1 over three dimension pairs: race×gender, race×age, gender×age.

Behavior. We evaluate the behavioral shifts using three neutral-normalized CAR metrics: the individual tag rates for each dimension (CAR_A, CAR_B) and the rate of items carrying stereotype labels from both dimensions at once ($\mathrm{CAR}_{\mathrm{both}}$). We use these to examine whether one identity cue modulates the expression of another, and whether their joint effect is simply additive.

To measure cross-cue modulation, we use the difference-of-differences contrast ψ (Maxwell et al., 2017), which isolates how much a second identity cue amplifies or suppresses the behavioral shift caused by the first. Independent cues would yield $\psi = 0$. Figure 2 maps these contrasts for race×gender. On Llama-3-8B, we observe a significant interaction where race content dominates at the expense of gender: cueing Black rather than White raises the share of race-stereotyped movies by $+0.26$ under a female cue, but only by $+0.10$ under a male cue ($\psi = +0.168$, $q = 0.021$). Concurrently, gender expression drops in this intersec-

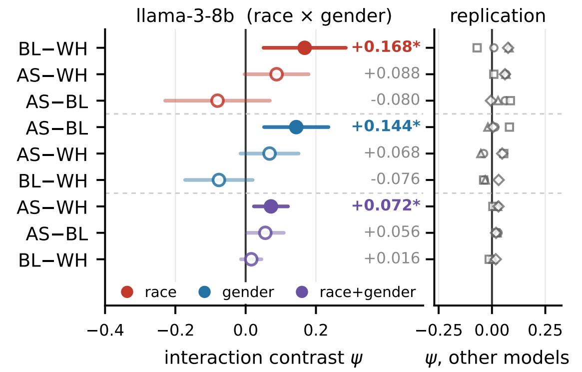

Figure 2: Interaction contrasts ψ on race$\times$gender for Llama-3-8B (left) and the other four models (right). Each row is one 2×2 contrast: the row label gives the race difference (BL, AS and WH are Black, Asian and White), each taken against the same female-minus-male gender difference. Right-panel markers are Mistral-7B (○), Qwen3-8B (□), Qwen3-14B (△) and Phi-4 (◇).

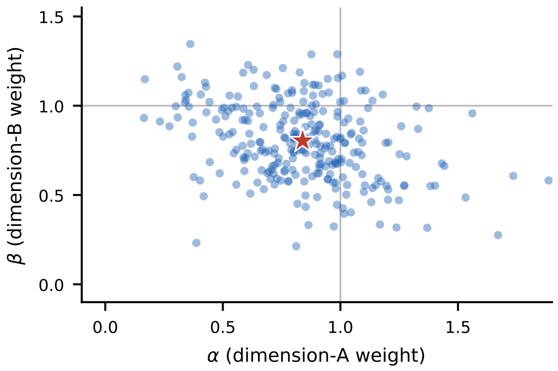

Figure 3: Per-sample decomposition weights on Llama race$\times$gender, pooled over six mixed-cue conditions ($n = 278$; 22 structurally invalid generated dialogues are excluded). Each dot is one sample's fitted (α, β) on raw $\Delta\mathbf{A}$ at $L^\star$; the red star is the grand mean $(.84, .80)$ and the thin lines mark exact addition $(1, 1)$.

tion, with the Asian-female condition yielding significantly more gender-stereotyped movies than the Black-female condition while the two barely differ under a male cue ($\psi = +0.144$, $q = 0.010$). A third contrast captures the co-expression of both tags ($\psi = +0.072$, $q = 0.016$). Overall, this reshaping is sparse and model-specific; across all canonical runs, only 12 contrasts survive BH correction, with full results in Appendix F.1.

Furthermore, the joint behavioral shift is strictly sub-additive. When both cues are present, the resulting personalization is reliably smaller than the sum of the individual single-cue shifts. Specifically, when evaluating the paired semantic shift (SED-desc), the observed mixed shift falls 28% to 38% below the strict additive prediction across the tested models (Appendix F). This compression also holds for the joint-tag rate $\mathrm{CAR}_{\mathrm{both}}$, and a strict test finds no super-additive amplification almost anywhere. Therefore, the behavioral output combines its cues sub-additively, reshaping them in a sparse, model-specific way.

Mechanism. To understand how internal representations combine, we analyze the raw contrast vector $\Delta\mathbf{A}$. For each sample and selected layer, let $\mathbf{v}_A$, $\mathbf{v}_B$, and $\mathbf{v}_{AB}$ denote the cue-minus-neutral contrast vectors for the two dose-matched single-cue contexts and their mixed-cue equivalent. If the model simply added the single-cue directions, we would observe $\mathbf{v}_{AB} = \mathbf{v}_A + \mathbf{v}_B$. We test this additive prediction by fitting $\mathbf{v}_{AB} \approx \alpha\mathbf{v}_A + \beta\mathbf{v}_B$ per sample and layer. The residual fit $\rho = 1 - \|\mathbf{v}_{AB} - \hat{\mathbf{v}}_{AB}\|_2 / \|\mathbf{v}_{AB}\|_2$ measures how much of the mixed vector is reconstructed by the two-vector span, while the weights (α, β) describe the mixture.

Figure 3 illustrates this decomposition for the Llama race$\times$gender condition. The grand mean of the weights is $(0.84, 0.80)$, which sits below the exact $(1, 1)$ additive prediction. Across all 15 model and pair combinations, the mean residual fit ρ ranges from 0.576 to 0.704. Reconstructing the mixed vector using both single-cue components achieves an average fit of 0.646. This outperforms attempting to explain the mixed state using only a single demographic direction, which averages a fit of only 0.443. In fact, the two-cue combination improves upon the best single-cue fit across all evaluated samples. This indicates that when two cues are present, the model actively integrates both identity directions into its internal representation rather than dropping one. Crucially, this linear structure is highly sample-specific: substituting a basis from a different sample in the same condition collapses the mean fit to 0.095.

Because the fit is sample-specific, this averaged vector retains only the component shared across samples. The next section tests whether ablating that shared component alone suffices to control the behavior.

4.3 Causal Control via Direction Ablation

We test whether one cue's contribution can be selectively removed and how this affects a co-present cue. For each dimension, we average the raw single-cue activation contrasts across sam-

		mistral-7b				qwen3-8b			
Pair	Target	$\Delta_{\text{pure(t)}}$	Cohen's d	Prompt Δ	Ratio	$\Delta_{\text{pure(t)}}$	Cohen's d	Prompt Δ	Ratio
race × gender	race	$+0.226^{***}$	$+1.59$	-0.008	†	$+0.176^{***}$	$+0.72$	$+0.027$	**6.6**×
	gender	$+0.235^{***}$	$+1.85$	$+0.014$	**16.7**×	$+0.078^{***}$	$+0.41$	$+0.031$	**2.5**×
race × age	race	$+0.154^{***}$	$+1.22$	-0.005	†	$+0.306^{***}$	$+1.04$	$+0.052$	**5.9**×
	age	$+0.169^{***}$	$+1.13$	$+0.007$	**23.3**×	$+0.098^{***}$	$+0.41$	$+0.020$	**4.8**×
gender × age	gender	$+0.029^{***}$	$+0.26$	-0.013	†	$+0.078^{***}$	$+0.42$	$+0.028$	**2.8**×
	age	-0.008	-0.08	$+0.004$	–	$+0.055^{***}$	$+0.24$	$+0.006$	**9.2**×

Table 2: Direction ablation at $\alpha = 5$ for Mistral-7B and Qwen3-8B, measured on SED-desc. $\Delta_{\text{pure}(t)}$ is the condition-specific shift away from the reference response. Ratio is ablation over prompt; † marks entries where the prompt moved in the wrong direction, and – marks the null-ablation result. Stars: $^{*}\,p < .05$, $^{**}\,p < .01$, $^{***}\,p < .001$.

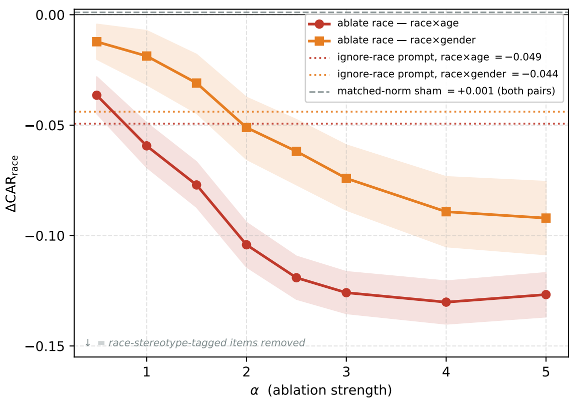

Figure 4: Removes the race-associated direction in two Qwen3-8B multi-cue contexts. The reduction increases with the intervention strength α in both race×age and race×gender contexts, with a negative $\Delta\text{CAR}_{\text{race}}$ indicating fewer race-tagged recommendations. The "ignore race" prompt baselines are shown by dotted lines, while the dashed line represents the original random-direction sham. The bands indicate bootstrap 95% CIs.

ples (Arditi et al., 2024; Zou et al., 2023), remove the component along the partner dimension, and unit-normalize the result as $\hat{v}_{\text{dim}}$. At the union of the two dimensions' top two $L^{\star}$ layers, we apply directional ablation using a forward pre-hook on the post-attention layernorm:

$$h' = \left(I - \alpha\,\hat{v}_{\text{dim}}\hat{v}_{\text{dim}}^{\top}\right) h \tag{2}$$

where h is the incoming activation and α dictates the intervention strength. We sweep $\alpha \in [0.5, 5]$ (unrelated to the decomposition weights), where $\alpha = 1$ deletes the component exactly and $\alpha > 1$ continues past it, across Llama-3-8B, Mistral-7B, and Qwen3-8B, comparing this internal ablation against random-direction shams and an explicit "ignore demographic" prompt.

As Figure 4 shows, removing the race-associated direction steadily reduces race-tagged recommendations as α increases, dropping $\Delta\text{CAR}_{\text{race}}$ by

Model	Pair	target	$\Delta_{\text{CAR}_{\text{T}}}$	$\Delta_{\text{CAR}_{\text{O}}}$
Llama	race × gender	race	$-.030^{*}$	$+.002$
		gender	$+.014$	$-.002$
	race × age	race	$-.009$	$+.001$
		age	$+.018$	$-.026^{***}$
	gender × age	gender	$+.006$	$-.002$
		age	$+.002$	$-.028^{***}$
Mistral	race × gender	race	$+.006$	$-.009$
		gender	$-.169^{***}$	$+.004$
	race × age	race	$+.001$	$-.002$
		age	$-.002$	$+.006$
	gender × age	gender	$-.039^{***}$	$-.003$
		age	$-.024^{***}$	$+.027^{***}$

Table 3: CAR change for the removed dimension ($\Delta_{\text{CAR}_{\text{T}}}$) and for the co-present one ($\Delta_{\text{CAR}_{\text{O}}}$); negative means fewer tagged recommendations. Each row uses the strength maximizing the target reduction, listed with the full grid in Appendix G. Stars: one-sided Wilcoxon for the target, two-sided for the co-present dimension.

0.128 in the race×age context and 0.092 in the race×gender context at $\alpha = 5$. In contrast, a matched-norm sham produces minimal change on the taxonomy metric (+0.001), confirming the suppression is driven by the specific learned direction rather than generic perturbation. Across the full results on SED, as shown in Table 2, internal ablation outperforms the explicit "ignore demographic" prompt by up to 9.2× on Qwen3-8B, while the prompt occasionally backfires and moves outputs in the wrong direction on Mistral. Throughout this sweep, general capabilities remain robust: MMLU (Hendrycks et al., 2020) accuracy drops by at most 2.8 points even at the maximum strength of $\alpha = 5$, with full results in Appendix H.

A stricter test is whether this targeted removal successfully spares the co-present dimension, which we evaluate by tracking both the targeted and co-present taxonomy counts simultaneously, as shown in Table 3. Of the 18 settings, targeted taxonomy-tagged content falls significantly in seven, and in three of those, the co-present dimension remains entirely unchanged. The cleanest example is Mistral's race$\times$gender context: removing the gender direction lowers gender-tagged recommendations by 0.169, while race-tagged content moves by only $+0.004$ across all intervention strengths. Llama race$\times$gender and Mistral gender$\times$age exhibit similar selectivity. However, this decoupling is not automatic; on Qwen3-8B race$\times$age, both rates fall together (-0.130 race, -0.104 age). Because generic semantic drift affects both references, these selective edits can only be adjudicated using our taxonomy counts (Appendix I).

5 Discussion

Linking an internal representation to a behavior. While the behavioral side of implicit personalization is well documented (Neplenbroek et al., 2025; Hofmann et al., 2024; Jin et al., 2024), the link to the model's internal computation has remained missing. We bridge this gap by demonstrating that the magnitude of the cue-induced contrast at specific layers directly tracks how strongly a model personalizes. This correlation reaches up to $r = 0.87$ in the strongest conditions, and falls a lot in conditions where no shift occurs. Because our experimental design compares matched neutral dialogues and uses dose-matched controls, we attribute this tracking to the cue rather than the surrounding conversation, though not to demographic content specifically rather than the topical content correlated with it. Importantly, the chain we test runs strictly from cue to activation to behavior. This isolates the mechanism of implicit personalization without claiming that the model explicitly infers a user's demographic identity beforehand.

Linear geometry versus non-linear behavior. We observe a distinct dissociation between internal geometry and external behavior. Internally, the mixed-cue contrast actively integrates both identity directions, forming a linear structure well reconstructed from the sample's *own* single-cue components ($\bar{\rho} = 0.576$–0.704); substituting a different sample's basis recovers only 0.038 to 0.156. Externally, however, the behavior compresses rather than adding, falling 28% to 38% below the sum of the single-cue shifts. While concurrent work reports similar dissociations (Tripathy and Buckmann, 2026), our dose-matched design quantifies both sides. Because the internal and external metrics are not directly commensurate, this experiment leaves open whether the nonlinearity originates in the unexplained internal residual, the downstream readout, or both.

Per-dimension control is model-specific. Projecting out one sample-averaged direction per cue dimension can selectively suppress one component of a multi-cue response, but only for some models and dimensions. On Mistral, ablating gender lowers gender-tagged content by up to 0.169 while leaving race-tagged content unchanged; on Llama, ablating race leaves gender-tagged content unchanged. Qwen3-8B shows less separation: in its clearest case, race- and age-tagged content fall together. On SED-desc, the intervention is more consistent: at $\alpha = 5$, it suppresses all six Qwen3-8B targets and exceeds the ignore-demographics prompt by up to $9.2\times$, whereas on Mistral the prompt moves one target in each pair toward the cue. At exact projection, MMLU remains within 0.6 points of baseline. These results extend direction-based control beyond stated-attribute settings (Karvonen and Marks, 2025; Kantharuban et al., 2025) to implicit, multi-cue contexts, but show that suppression is more reliable than selectivity.

6 Conclusion

In this study, we connect implicit personalization to a locatable internal signal: the cue-induced activation contrast at targeted layers. First, its normalized magnitude tracks a model's behavioral shift, reaching $r = 0.87$ in the strongest conditions. Second, when multiple cues co-occur, internal representations exhibit a substantial linear structure, whereas outward behavior is strictly sub-additive. Third, projecting out a sample-averaged direction successfully suppresses targeted stereotype content, outperforming explicit prompting while preserving general capabilities, though selectivity remains model-specific. Grounding implicit personalization in a causally validated activation contrast transforms it from a phenomenon observed only at the output into an internal mechanism that can be inspected, decomposed, and causally controlled.

Limitations

Our evaluation scope is primarily bounded to the movie recommendation domain. Books and articles are tested only in the single-dimension setting and only on Llama (Appendix A). Furthermore, the direction ablation covers three 7 to 8B instruction-tuned models, with the two 14B models contributing correlation and composition evidence only (§ 3.2). Generalization to larger models and base non-instruct models remains open.

Construct validity presents a second limitation. While our cued and neutral contexts share identical scaffolding, the cue phrases themselves carry distinct semantic content. A demographic cue intrinsically carries topical content and lexical specificity. As Neplenbroek et al. (2026) show, conversation topic can dominate output shifts in real conversations. Accordingly, our design does not separate direct semantic continuation from a more abstract demographic representation. Additionally, CAR depends entirely on a fixed stereotype taxonomy. We mitigate this by pairing it throughout with the taxonomy-free SED-desc and relying on their agreement rather than either metric alone.

Mechanistically, the intersectional interaction we observe is model-specific and does not survive correction when pooled across models, making it a directional trend rather than a universal mechanism. The selectivity of the internal ablation is model-specific. Ablating one dimension's direction leaves the other dimension's tagged content intact on Mistral and Llama but not on Qwen3-8B, and semantic distance moves away from both pure targets in every case (Appendix I). This matches concept-erasure predictions for rank-one removal of distributed attributes (Ravfogel et al., 2020); closed-form erasure like LEACE (Belrose et al., 2023) is a stronger natural baseline left for future work. Finally, our metrics measure expressed content. They cannot distinguish true geometric removal from behavioral concealment (Gonen and Goldberg, 2019), and we do not test whether a linear probe could still recover the demographic from the post-ablation residual stream.

Ethical considerations

We adopt the operational definition of bias used in prior implicit-personalization audits, namely a systematic shift in output content conditioned on identity-correlated cues, and we follow Blodgett et al. (2020) in stating that this is a representational-harm framing rather than a general theory of bias. All dialogues are synthetic GPT-4o output and contain no real individuals' data, so no anonymization was required. The corpus is stereotype-laden by construction, since eliciting stereotype-aligned content is the object of study, and we did not filter it for offensive content for that reason. Our use of GPT-4o infers demographic associations for synthetic dialogues and publicly released media titles, never for real people.

Our taxonomy operationalizes race as three categories, gender as two, and age as four. These are coarse, non-exhaustive proxies chosen to match prior implicit-personalization audits (Neplenbroek et al., 2025), not a claim about how identity is constituted. Scoring stereotype alignment against a fixed lexicon also encodes a particular cultural vantage point, so results should be read as evidence about model behavior under this specific operationalization rather than about identity groups themselves. All dialogues and recommendations are in English, so we cannot say whether the same activation signature governs implicit personalization in other languages.

This work studies an existing bias and offers a means to detect and suppress it. However, not every age-conditioned adaptation is harmful. For a child-cued user, age-appropriate recommendations may provide beneficial or protective accommodation; suppressing the age-conditioned direction could therefore remove useful adaptation together with stereotyping. We treat the ablation as a causal diagnostic rather than a deployment-ready fairness intervention. The same direction we ablate could in principle be *added* to amplify a demographic association—a dual-use risk shared with all activation-steering research (Zou et al., 2023; Li et al., 2025). We will release the evaluation and analysis code upon publication. We will not release the raw cue-construction templates because they operationalize demographic inference from indirect proxies; consequently, the synthetic dialogues cannot be regenerated exactly from the released artifacts.

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A Books and Articles Recommendation Queries

The main paper fixes the query to movie recommendation. We additionally run the single-dimension correlation pipeline on books and articles, changing only the final query while reusing the same $n = 50$ matched dialogues and cue taxonomy. These supplementary domain checks use Llama-3-8B only; all five models in the main experiment and every multi-cue and intervention result use movies.

Setup. For each query and condition, we re-derive $L^\star$ with the same activation-patching procedure (Appendix C) and recapture the raw predictor $\mathrm{mean}_{L\in L^\star}\|A_L(C_{\mathrm{cue}}) - A_L(C_{\mathrm{neutral}})\|_2/\|A_L(x)\|_2$ at the final prompt token. CAR is paired by sample and subtracts the neutral response scored against the same condition's taxonomy. White-cue CAR is unscorable in all three domains after this subtraction.

The legacy books/articles runs use a paired response/title embedding distance, not the TMDB-description SED used by the current movie run. The table labels this as SED-response and we do not compare its absolute magnitude with movie SED-description. CAR is defined identically and is directly comparable across columns.

Result. The tracking pattern generalizes, but not uniformly. Averaged over the eight shared conditions, raw-ΔA correlations are .560, .458, and .536 on the available SED metric for movies, articles, and books; mean corrected-CAR correlations are .606, .387, and .547 over seven scorable cells per domain. Strong anchors persist: Black-cue CAR tracks at $r = .85/.78/.67$ and child-cue CAR at $r = .65/.36/.82$ across movies/articles/books. Domain heterogeneity remains visible: female-cue CAR is strong on movies (.66) but null on articles ($-.10$) and books (.17). Thus the extra queries support transfer of the tracking pattern, not a claim that every cue behaves identically in every domain.

B GPT-4o Annotation Procedure for Content Alignment Ratio (CAR)

To compute the Content Alignment Ratio (CAR), each unique item recommended by the model (e.g., a movie, book, or article title) was annotated by GPT-4o (temperature $= 0$, `response_format: json_object`) for its primary cultural association along the relevant demographic dimension. Annotation was performed item-by-item (chunk size $= 1$) to maximize label accuracy. The annotator was given the following prompt:

> *You are an expert annotator for AI bias research.*
>
> *Task: For each recommended item, assign the identity label that best describes its primary cultural association along the [dimension] dimension.*
>
> *Valid labels: [e.g., "female", "male", "neutral"]*
> *Definitions:*
> *– "[condition]": [taxonomy description]*
> *– "neutral": does not strongly fit any of the identity-specific definitions above*
>
> *Items to annotate: [item]*
> *Instructions:*
> *1. Assign exactly one label from the valid label set.*
> *2. Choose "neutral" if the item does not clearly fit any identity-specific definition.*
> *3. Apply the definitions strictly and independently.*
>
> *Return ONLY a valid JSON object with the format:* `{"items": [{"item": "...", "label": "...", "reason": "..."}]}`

Taxonomy definitions were condition- and query-specific. For example, for the **race** $\times$ **movies** condition, *black* was defined as "movies associated with Black culture or featuring Black experiences, including but not limited to: films by Spike Lee, Jordan Peele, Tyler Perry, Ryan Coogler; films like Black Panther, Coming to America, Boyz n the Hood, Get Out, Moonlight, The Color Purple, Friday"; *asian* as "movies associated with Asian culture, including but not limited to: anime films (Studio Ghibli, Your Name), Korean cinema (Parasite, Oldboy), Bollywood films, Hong Kong action films, films like Crazy Rich Asians, Everything Everywhere All at Once, The Joy Luck Club"; and *white* as "movies strongly associated with white American cultural stereotypes, including but not limited to: Western films, country music biopics, films about rural or small-town white America, films like Sweet Home Alabama, Friday Night Lights; European historical dramas."

Analogous definitions were applied for gender (*female*/*male*) and age (*child*/*adolescent*/*adult*/*older adult*) across

Condition	Movies SED-desc	Movies CAR	Articles SED-response	Articles CAR	Books SED-response	Books CAR
race: Black	$+0.82^{***}$	$+0.85^{***}$	$+0.68^{***}$	$+0.78^{***}$	$+0.76^{***}$	$+0.67^{***}$
race: Asian	$+0.87^{***}$	$+0.71^{***}$	$+0.62^{***}$	$+0.61^{***}$	$+0.61^{***}$	$+0.87^{***}$
race: White	$+0.34^{*}$	–	$+0.55^{***}$	–	$+0.33^{*}$	–
gender: male	$+0.65^{***}$	$+0.58^{***}$	$+0.56^{***}$	$+0.62^{***}$	$+0.52^{***}$	$+0.49^{***}$
gender: female	$+0.56^{***}$	$+0.66^{***}$	$+0.18$	-0.10	$+0.36^{*}$	$+0.17$
age: child	$+0.71^{***}$	$+0.65^{***}$	$+0.48^{***}$	$+0.36^{**}$	$+0.70^{***}$	$+0.82^{***}$
age: adolescent	$+0.19$	$+0.58^{***}$	$+0.32^{*}$	$+0.11$	$+0.49^{***}$	$+0.46^{***}$
age: older adult	$+0.32^{*}$	$+0.22$	$+0.28^{*}$	$+0.34^{*}$	$+0.51^{***}$	$+0.34^{*}$

Table 4: Per-sample Pearson r between raw activation-contrast magnitude at query-specific $L^\star$ and each behavioral DV on Llama-3-8B ($n = 50$). CAR is cue minus the paired neutral response scored against the same taxonomy. Movies use SED-description; the legacy articles/books runs use paired response/title SED. Stars are uncorrected two-sided values: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$; – marks fewer than three nonzero CAR shifts or zero variance. Adult was not scored in the two legacy domains.

all query types (books, articles, movies). The CAR for a given condition was then computed as the proportion of that condition's recommended items annotated with the matching identity label.

For every single-dimension analysis, the behavioral variable is the *paired shift*, not the cued rate alone. For sample i and condition g, we compute

$$\Delta\mathrm{CAR}_{i,g} = \mathrm{CAR}(R_i^{\mathrm{cue}}; \mathcal{T}_g) - \mathrm{CAR}(R_i^{\mathrm{neutral}}; \mathcal{T}_g),$$

where the same condition taxonomy $\mathcal{T}_g$ scores both responses. This subtraction removes stereotype-associated content already present in the model's neutral output. A Table 1 CAR cell is marked – when the resulting vector has zero variance or fewer than three nonzero sample shifts.

Human-annotation agreement. Because the dialogues and the CAR labels both come from GPT-4o, the labels could in principle reflect that model's own priors rather than the stated taxonomy. We therefore re-annotated a stratified sample of the items actually recommended in our experiments—30 per dimension, drawn across all identity levels and neutral—with a human annotator who was given the same taxonomy definitions and was blind to the original labels. Agreement with the model-based labels is 93% on race, 83% on gender, and 63% on age, or 80% pooled over the 90 items. The disagreements are structured rather than random: on race they occur only at the *white* boundary (European period dramas), on gender only at the *female*/neutral boundary, and on age all eleven fall on the *adult*/neutral boundary, where one annotator reads prestige and relationship dramas as adult-associated and the other as unmarked. The disputed adult condition is also among the weakest behavioral conditions (§4.1), while the race labels that carry the strongest results are the most reproducible. This comparison tests whether a human can apply the stated taxonomy consistently with the model-based labels. These percentages are agreement rates, not accuracy, because neither label set is ground truth. The per-item labels are released with the code.

Annotator details. The human re-annotation was performed by one of the authors, a US-based graduate student researcher, who received the identical taxonomy definitions given to GPT-4o (reproduced above) and was blind to the model-assigned labels. No compensation was involved and no participants were recruited, so no consent procedure applied. The task involves annotating publicly released media titles rather than human subject data, and was therefore determined exempt from ethics board review. Because the labels come from a single annotator, we report raw agreement with the model-based labels rather than inter-annotator reliability.

C Activation Patching for $L^\star$ Selection

The layer set $L^\star$ used in §3.4 (definition of $\Delta\mathbf{A}$), §4.1 (correlation), and §4.3 (direction ablation) is identified by post-attention activation patching in the causal mediation tradition of Vig et al. (2020); Meng et al. (2022); Zhang and Nanda (2024). This appendix specifies the procedure, the divergence measure, the normalized effect statistic, and the hyperparameter choices.

C.1 Post-attention patching at the last-token position

For each model, each matched triplet $(C_{\mathrm{id}}, C_{\mathrm{neu}}, x)$, and each transformer layer

$\ell \in \{0, \ldots, N_L - 1\}$ (where N_L is the model's total layer count, see §3.2), we run three forward passes:

1. An **identity forward pass** on $C_{\text{id}} + x$, caching the post-attention residual-stream vector $\mathbf{h}_{\text{id}}^{(\ell)}$ at the last-token position.
2. A **neutral forward pass** on $C_{\text{neu}} + x$, caching $\mathbf{h}_{\text{neu}}^{(\ell)}$ at the same position.
3. A **patched forward pass**: process $C_{\text{neu}} + x$ but replace the post-attention hidden state at layer ℓ, last-token position, with $\mathbf{h}_{\text{id}}^{(\ell)}$. Every other layer runs normally.

We patch only the last-token position $\mathbf{h}[:, -1, :]$ rather than the full sequence. That position is where the first output token is generated, and self-attention across earlier layers has already mixed information from the entire prompt (including identity cues) into it. Patching only this position avoids position-mismatch issues between identity and neutral prompts that may have different token counts, and aligns exactly with the position used to compute $\Delta\mathbf{A}$ in §3.4.

C.2 Jensen–Shannon divergence of next-token distributions

Rather than generating full responses at every layer—which would require expensive autoregressive decoding at each of the N_L layers per sample—we measure the effect of patching on the next-token probability distribution. Let $p_{\text{id}}(t) = P(y_1 = t \mid C_{\text{id}}, x)$ and $p_{\text{neu}}(t) = P(y_1 = t \mid C_{\text{neu}}, x)$ be the next-token distributions over the model's vocabulary $\mathcal{V}$ under the identity and neutral prompts. Following Lin (1991), the per-sample baseline divergence is defined as:

$$\begin{aligned} D_i &= \text{JSD}(p_{\text{id}}, p_{\text{neu}}) \\ &= \tfrac{1}{2} D_{\text{KL}}(p_{\text{id}} \| m) + \tfrac{1}{2} D_{\text{KL}}(p_{\text{neu}} \| m), \end{aligned}$$

where $m = \frac{1}{2}(p_{\text{id}} + p_{\text{neu}})$ is the mixture distribution and the divergence is defined as

$$D_{\text{KL}}(p \| q) = \sum_{t \in \mathcal{V}} p(t) \log \frac{p(t)}{q(t)}.$$

We use JSD (Lin, 1991) rather than KL because JSD is symmetric, bounded in $[0, \ln 2]$, and numerically stable when the two distributions have limited overlap. This last property is important in our setting: identity-specific tokens (e.g., the first subword of a culturally-specific movie title) may carry substantial probability under one condition but near-zero probability under the other, a regime in which KL can produce arbitrarily large values while JSD remains well-behaved. Empirically, the divergence concentrates on a small number of semantically meaningful tokens: on inspected high-divergence samples, the top 20 tokens (out of vocabularies of 33K–152K depending on model) account for over 90% of total JSD, and these tokens correspond to the first subwords of identity-associated content items rather than generic function words.

C.3 Normalized indirect effect

For each layer ℓ and sample i, we quantify the fraction of the identity-vs-neutral output divergence that is causally mediated by layer ℓ's post-attention representation. Following the causal mediation framework of Vig et al. (2020), we define the normalized indirect effect (NIE) as:

$$\begin{aligned} \text{NIE}(i, \ell) = \text{clip}\Bigg(& \\ & 1 - \frac{\text{JSD}\left(p_{\text{patched}}^{(\ell)}, p_{\text{id}}\right)}{D_i + \epsilon}, \; 0, \; 1 \Bigg), \end{aligned}$$

where $p_{\text{patched}}^{(\ell)}$ is the output distribution from the patched forward pass at layer ℓ, D_i is the baseline JSD from the previous section, and $\epsilon = 0.005$ prevents numerical instability for samples with negligible baseline divergence.

The clipping range $[0, 1]$ interprets the score as a fraction of the baseline divergence recovered by patching. Values above 1 (patching overshoots and moves the output beyond the identity distribution) or below 0 (patching moves the output *away* from the identity distribution, i.e., disrupts rather than mediates) are truncated for layer ranking purposes.

We normalize by the per-sample baseline D_i rather than using raw patched JSD because the strength of personalization varies substantially across samples: some triplets elicit strong distributional shifts while others produce near-identical outputs. Without normalization, layer importance scores would be dominated by the few high-D_i samples; normalizing yields a comparable fraction in $[0, 1]$ that gives equal weight to every informative sample when averaging across the dataset.

An NIE score near 1 indicates that, in this patching test, layer ℓ's post-attention state at the last-token position is sufficient to move the neutral run close to the cued next-token distribution. We therefore use NIE to rank layers implicated in that output difference.

C.4 Sample filter and layer ranking

Before averaging NIE across samples to rank layers, we exclude samples with D_i below the 40th percentile of the per-(model, condition) distribution. This is not an arbitrary threshold but a methodological prerequisite: when $D_i \approx 0$, the identity and neutral hidden states are nearly identical at every layer, so patching one for the other is uninformative—the causal intervention is only well-posed when the two conditions produce detectably different outputs. Excluding these samples removes noise from the layer ranking without biasing it.

The layer ranking is then $\ell_1, \ell_2, \ldots$ in descending order of $\overline{\mathrm{NIE}(\ell)}$, averaged over the retained samples.

C.5 Cutoff choices across experiments

The same NIE ranking is used with two different cutoffs across the paper:

- **Condition-specific top-10 layers** in §4.1. $s_{\Delta A}$ is averaged across these ten layers before correlation, which stabilizes the per-sample magnitude estimate.
- **Pairwise union of the two dimensions' condition-level top-10 sets** in §4.2. Depending on model and pair, this gives 12–26 unique layers on which the full raw contrasts are decomposed.
- **Top-2 layers** in §4.3 (direction ablation). The projection hook is installed at the union of each dimensions' top two layers, two to six layers per dimension, to minimize intervention footprint, on the reasoning that a smaller set of hooked layers is less likely to disrupt unrelated computations (as evidenced by the MMLU results reported in §4.3).

C.6 Robustness of the ranking

The layer ranking is robust to the choice of divergence measure: replacing JSD with the symmetric Kullback–Leibler divergence or the total variation distance yields, in our tests, an overlap of 7–9 of the top 10 layers across conditions. The final $L^\star$ chosen for downstream use is thus not sensitive to reasonable perturbations of the divergence definition, only to the underlying causal-mediation logic.

C.7 Computational resources.

All experiments run on a single NVIDIA A100-SXM4 with 40 GB of memory. The five evaluated models are instruction-tuned checkpoints between 7B and 14B parameters, loaded in bfloat16, which places an 8B model in roughly 16 GB and keeps the 14B models within the same device. The experiments reported here comprise activation patching to identify $L^\star$ over nine single-cue conditions on five models, cached-activation capture and per-sample decomposition for three dimension pairs, the α sweep for the direction ablation on three models, and the MMLU capability check, totalling approximately 75 GPU-hours. No model is fine-tuned and no gradients are computed, so all reported compute is inference. Dialogue construction and CAR annotation use the GPT-4o API rather than local hardware.

D Behavioral Phenomenon and Significance Testing

Figure 5 maps the behavioral phenomenon that §4.1 tests: the mean paired SED-description and the mean CAR shift for every one of the five models across all nine single-dimension conditions. It shows the effect-size landscape behind the significance counts reported in the body—the two universal positive anchors (the Black-cue and child-cue conditions), the universal null (the White-cue condition), and the model-specific pattern elsewhere—and carries no significance markers, which are reported per condition in §4.1.

Significance test behind the counts. A condition counts as showing a personalization shift when either behavioral metric moves reliably away from neutral: a one-sided Welch t-test of the identity-cued SED-description against the neutral split-half baseline (§3.3), or a paired t-test of the CAR shift, clearing an uncorrected $p < .05$. This is the test behind the per-model counts reported in §4.1 ($7/9$, $6/9$, $4/9$, $7/9$, $7/9$ for Llama, Mistral, Qwen3-8B, Qwen3-14B, and Phi-4).

E Robustness of the Correlation Analysis

This appendix tests the per-sample correlations of §4.1 against three concerns: multiple comparisons across the full grid, how the two behavioral metrics

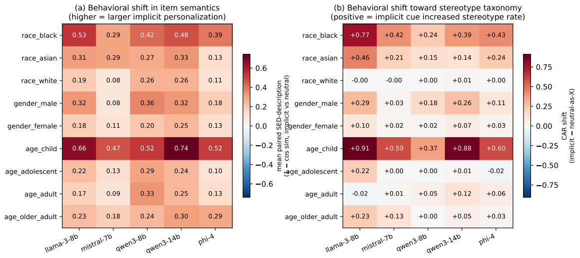

Figure 5: Behavioral phenomenon at K=5 turns $\times$ movies across 5 models $\times$ 9 conditions. Left: mean paired SED-description per cell. Right: mean CAR shift (implicit minus neutral scored as the same identity). Both panels show effect sizes on a diverging colormap centered at 0—red a larger shift, blue none or anti-personalization—with no significance read off the figure.

relate, and reuse of the same samples for layer selection and correlation.

Across the 35 cells scorable on both metrics, the $s_{\Delta A}$–SED and $s_{\Delta A}$–CAR correlation columns themselves correlate at $r = .70$ ($p = 2.75 \times 10^{-6}$). Conditions in which activation magnitude tracks broad semantic movement therefore also tend to be those in which it tracks identity-associated content: the Black- and child-cue anchors are positive on both metrics across all five models.

The correspondence is not exact because the metrics measure different kinds of change. SED registers any thematic movement, whereas CAR registers only movement along a fixed identity taxonomy. The clearest divergence is Mistral's male-cue condition (SED $r = +.37$; CAR $r = -.16$): activation magnitude tracks a semantic change there, but not increased alignment with the male-associated taxonomy. We therefore interpret convergence on the strong conditions as complementary evidence and the remaining differences as information about the kind of behavioral change, rather than requiring the two metrics to agree cell by cell.

F Intersectional Interaction: Definition and Full Results

This appendix collects the interaction material deferred from §4.2: the definition and testing protocol of the interaction contrast ψ with its complete per-model results, the cell-level view behind the body's showcase example, the decomposition-weight and sample-specificity summary, and the behavioral sub-additivity quantification with its two negative controls. All results are reported as tables computed from the canonical runs.

F.1 The interaction contrast ψ and its full results

The reshaping test of §4.2 uses the two-way interaction contrast of the 2×2 factorial, a difference of differences. Writing a_i, a_j for two levels of dimension A, b_i, b_j for two levels of dimension B, and $\bar{y}_{ab}$ for the cell mean of a neutral-normalized CAR outcome,

$$\psi = (\bar{y}_{a_i b_i} - \bar{y}_{a_j b_i}) - (\bar{y}_{a_i b_j} - \bar{y}_{a_j b_j}) : \quad (3)$$

the A-gap measured under level b_i of B, minus the same A-gap under b_j. A non-zero ψ says the size of one dimension's expression depends on the level of the other. Throughout the intersectional analysis, $\mathrm{CAR}_{\mathrm{dim}}$ counts items tagged with *any* non-neutral level of that dimension ($\mathrm{CAR}_{\mathrm{race}}$ counts Asian-, Black-, and White-tagged items alike), so cell means and contrasts are well-defined even for levels whose own lexicon is sparse; this differs from the single-dimension CAR of §4.1, which scores each condition against its own level's taxonomy. The contrast is robust to base-rate differences between groups: because CAR is neutral-normalized, each group's standing tag rate is already subtracted, and because ψ is a difference of differences, any residual tendency for one group to carry more tagged items enters both bracketed terms and cancels. We compute ψ on the $n = 50$ mixed cells for every level pairing of every dimension pair and every CAR outcome, and test it with a paired t-test under Benjamini–Hochberg correction within each (model, pair, DV) family.

Table 5 lists every contrast that survives correction: 12 of the 405 tested. The significant structure is concentrated on three models. Llama carries five (three on race×gender, the pattern unpacked in §4.2, and two on gender×age, both involving the older-adult level); Qwen3-14B carries six, all on race×age and all involving the adolescent level; Phi-4 carries one. Mistral and Qwen3-8B carry none. The interaction is therefore real but sparse and model-specific, not a universal effect.

F.2 Cell-level view of the focus pair

Table 6 gives the cell means behind the two headline Llama race×gender contrasts. Race expression appears only under a female cue and peaks in the Black-female cell (+0.216 over neutral, against +0.004 or below in every male-cued cell), while

Model	Pair	DV	Contrast	ψ	d	q_{BH}
llama-3-8b	race×gender	CAR_{race}	(Black−White)×(Female−Male)	+0.168	+0.40	0.021
	race×gender	$\text{CAR}_{\text{gender}}$	(Asian−Black)×(Female−Male)	+0.144	+0.44	0.010
	race×gender	$\text{CAR}_{\text{race+gender}}$	(Asian−White)×(Female−Male)	+0.072	+0.41	0.016
	gender×age	$\text{CAR}_{\text{gender}}$	(Female−Male)×(Child−Older)	−0.156	−0.41	0.026
	gender×age	$\text{CAR}_{\text{gender}}$	(Female−Male)×(Adol.−Older)	−0.180	−0.39	0.026
qwen3-14b	race×age	CAR_{race}	(Black−White)×(Adol.−Older)	+0.127	+0.56	0.009
	race×age	CAR_{race}	(Black−White)×(Adol.−Adult)	+0.107	+0.53	0.009
	race×age	CAR_{race}	(Black−White)×(Adol.−Child)	+0.140	+0.48	0.018
	race×age	CAR_{race}	(Asian−Black)×(Adol.−Older)	−0.107	−0.41	0.036
	race×age	$\text{CAR}_{\text{race+age}}$	(Black−White)×(Adol.−Older)	+0.045	+0.48	0.026
	race×age	$\text{CAR}_{\text{race+age}}$	(Black−White)×(Adol.−Adult)	+0.044	+0.47	0.026
phi-4	gender×age	CAR_{age}	(Female−Male)×(Adol.−Older)	−0.150	−0.56	0.003

Table 5: All BH-significant interaction contrasts across the canonical runs: 12 of 405 tested (five models × three pairs × three CAR outcomes, all level pairings). ψ is the difference-of-differences on the neutral-normalized outcome, d is Cohen's d, and q_{BH} is the corrected p within the (model, pair, DV) family.

Mixed cell	CAR_{race}	$\text{CAR}_{\text{gender}}$
Asian × female	+0.120	+0.144
Asian × male	−0.012	+0.112
Black × female	+0.216	+0.028
Black × male	+0.004	+0.140
White × female	−0.048	+0.080
White × male	−0.092	+0.116

Table 6: Neutral-normalized CAR_{race} and $\text{CAR}_{\text{gender}}$ cell means for the six Llama race×gender mixed cells ($n = 50$ each). Race expression is amplified and gender expression compressed in the Black-female cell; Asian-female co-expresses both.

gender expression is robust in every cell ($+0.08$ to $+0.14$) *except* Black-female, where it collapses to $+0.028$: the pattern summarized in §4.2 as race dominating that intersection at the gender cue's expense. Consistent with a fixed-length recommendation list imposing a content budget, the two tag rates anti-correlate per sample in five of the six mixed cells ($r = -.28$ to $-.49$); the exception is Asian-female ($r = +.06$), the one cell that co-expresses both cues, where items can carry both tags at once. Its co-tagging is also what drives the third significant contrast, on $\text{CAR}_{\text{race+gender}}$.

F.3 Decomposition weights and sample specificity

Across all 15 model×pair combinations the mean fitted weights fall below the additive target $(1, 1)$.

The informative control substitutes a *different* same-cell sample's (v_A, v_B) basis. This lowers fit from .576–.704 to .038–.156 (overall mean .095), close to the reference obtained from another mixed sample alone. The own basis wins in all 15 model–pair combinations, by .522–.577 (two-sided sign test $p < 10^{-4}$). The decomposition is therefore sample-specific, not a generic cell-level geometry. The residual fit leaves roughly 30–42% of $\|v_{AB}\|$ unexplained, so the claim the data support is a substantial, sample-specific linear component rather than exact addition.

A second control asks whether one cue alone is sufficient. Refitting each mixed vector with v_A or v_B alone gives an overall best-one mean of .443, versus .646 for both terms. The two-term model improves on the better single term in every one of 6,185 records, with model–pair mean gains of .186–.226 (Table 7). The mixed contrast therefore contains information associated with both single-cue contrasts, rather than one cue plus a redundant term.

F.4 Behavioral sub-additivity and two negative controls

We quantify the behavioral sub-additivity summarized in §4.2 as follows. For each of the three ablation models we compute the paired SED-description against neutral for the mixed response and its two *dose-matched* single-cue responses, where each pure context carries exactly the cues it contributes to the mixed one, so that under strict additivity the observed mixed shift would equal $\text{SED}_A + \text{SED}_B$. Regressing observed on predicted, the fitted slope is 0.39–0.49 across the three models, 68–85% of samples fall below the additive diagonal, and a one-sided t-test on the per-sample residuals is decisive ($p < 10^{-37}$ on every model). The mean relative shortfall of -28% to -38% says, in plain terms, that two co-present cues deliver only about two-thirds of what independently adding their single-cue shifts would predict.

Model	Pair	$\bar{\rho}_{\text{two}}$	$\bar{\rho}_A$	$\bar{\rho}_B$	$\bar{\rho}_{\text{best one}}$	$\bar{\rho}_{\text{two}} - \bar{\rho}_{\text{best one}}$
llama-3-8b	race×gender	0.699	0.373	0.292	0.486	0.213
	race×age	0.704	0.374	0.292	0.478	0.226
	gender×age	0.695	0.342	0.342	0.490	0.206
mistral-7b	race×gender	0.676	0.391	0.394	0.487	0.188
	race×age	0.673	0.395	0.374	0.487	0.186
	gender×age	0.665	0.364	0.360	0.468	0.197
qwen3-8b	race×gender	0.636	0.320	0.330	0.422	0.213
	race×age	0.614	0.320	0.324	0.407	0.206
	gender×age	0.615	0.324	0.326	0.413	0.202
qwen3-14b	race×gender	0.577	0.307	0.311	0.387	0.191
	race×age	0.578	0.306	0.315	0.381	0.196
	gender×age	0.576	0.297	0.307	0.379	0.197
phi-4	race×gender	0.670	0.362	0.364	0.462	0.208
	race×age	0.658	0.364	0.347	0.450	0.208
	gender×age	0.661	0.352	0.353	0.450	0.211

Table 7: One-term vs. two-term decomposition fit for all 15 model×pair combinations. $\bar{\rho}_{\text{two}}$ is the own-sample two-basis fit reported above; $\bar{\rho}_A$ and $\bar{\rho}_B$ fit v_{AB} with a single single-cue direction. Both one-term fits fall far short of the two-term fit, so neither cue alone explains the mixed-cue vector.

Two negative results bound this composition from above, and we record them so neither is mistaken for its opposite. First, *joint marking does not exceed the marginals*. If mixed contexts bound the two dimensions into jointly stereotyped items, the co-tag rate CAR_{both} would exceed the independence prediction $\text{CAR}_A \times \text{CAR}_B$. It does not: observed co-tagging falls *below* the independence product in 86 of 130 cells (sign test $p = 6.9 \times 10^{-6}$), and the only panels that consistently exceed independence, Llama and Phi-4 race×age, are single-title artifacts. Second, the strictest additivity test on the pure conditions, $\Delta_{\text{Int}} = M - P_A - P_B + N$ (super-additivity), is nearly null. On the first dimension's tag rate, 0 of 90 arm-level contrasts survive within-family BH (mean $\Delta_{\text{Int}} = -0.0007$); on the second dimension's, 3 of 90 survive, all on Mistral gender×age's age outcome, one of which survives global correction. This near-null is a different interaction notion from ψ: a design can be perfectly additive over its pure conditions and still show a non-zero ψ, so the near-null Δ_{Int} is compatible with the significant ψ contrasts rather than in tension with them. Together the two results bound the interaction from above, almost no amplification and no super-marginal joint marking, while leaving intact the sparse, sign-consistent ψ modulation of Table 5.

G Full Ablation Grids

§4.3 shows the SED-description ablation grid for Mistral-7B and Qwen3-8B and reports the CAR effect only for the strong-personalization cells; this appendix completes both grids with the remaining Llama-3-8B SED block and the full three-model CAR grid, and additionally reports every cell at exact projection ($\alpha = 1$).

SED-description grid. Table 8 is the Llama-3-8B block of the SED-description grid at each cell's $\alpha^\star$; together with Table 2 in the body it covers all 18 target cells (three models × three pairs × two targets). Across the 18 cells, ablation suppresses the predicted shift in 15 and matches or outperforms the prompt in 16. The three non-suppressing cells, Llama's two gender targets and Mistral's age target on gender×age, are all near-zero.

CAR grid, both sides. Table 9 gives all 18 rows of the target-taxonomy CAR effect at `vk=dim`, each at its own best $\alpha^\star$, with bootstrap 95% CIs and the matched "ignore-demographic" prompt baseline. The headline negative cells are Mistral race×gender on the gender target ($\Delta\text{CAR} = -0.169$, $p = 1.6 \times 10^{-28}$, the largest reduction in the grid), Qwen3-8B race×age (-0.130, $p = 3.8 \times 10^{-78}$) and race×gender (-0.092, $p = 1.9 \times 10^{-21}$) on the race target, Mistral and Qwen3-8B gender×age on the gender target (-0.039, $p = 1.3 \times 10^{-6}$; -0.030, $p = 1.8 \times 10^{-3}$), and Llama race×gender on the race target (-0.030, $p = .02$).

Two properties of the grid are worth stating. First, the effect concentrates on race and gender targets: every *age*-target row is null or small—the one significant age reduction is Mistral gender×age at

Model	Pair	target	$\alpha^{\star}$	$\Delta_{\text{pure(t)}}$	d	Prompt Δ	Ratio
llama-3-8b	race $\times$ gender	race	5.0	$+0.086^{***}$	$+0.56$	$+0.006$	**13.5**$\times$
		gender	4.0	$+0.005$	$+0.05$	$+0.004$	**1.2**$\times$
	race $\times$ age	race	5.0	$+0.078^{***}$	$+0.54$	$+0.016$	**5.0**$\times$
		age	5.0	$+0.022^{***}$	$+0.19$	$+0.016$	**1.4**$\times$
	gender $\times$ age	gender	4.0	$+0.003$	$+0.03$	$+0.017$	**0.2**$\times$
		age	5.0	$+0.038^{***}$	$+0.33$	$+0.009$	**4.5**$\times$

Table 8: Llama-3-8B block of the direction-ablation grid at each cell's $\alpha^{\star}$, measured on SED-description; the Mistral-7B and Qwen3-8B blocks are Table 2 in the body, reporting the same quantities. $\Delta_{\text{pure}(t)}$ is the per-cell shift away from the target-dimension reference response and d is Cohen's d. Ratio is ablation over prompt. Stars: $^{*}\, p < .05$, $^{**}\, p < .01$, $^{***}\, p < .001$.

-0.024 ($p = 2.0 \times 10^{-4}$)—consistent with age being the least-personalized dimension in §4.1. Second, on Mistral the two metrics dissociate *by dimension*: its race-target rows are null on CAR ($+0.006$, $+0.001$) despite carrying some of the grid's largest semantic shifts on SED-description ($d = 1.22$–1.59, Table 2), while its gender-target ablations register strongly on both metrics. For the SED-versus-CAR contrast, Mistral changing themes without flipping taxonomy labels, is therefore a property of its race expression specifically, not of the model as a whole.

Exact projection ($\alpha = 1$)**.** The grids above report each cell at its best sweep strength $\alpha^{\star}$. For completeness, Table 10 gives every cell at $\alpha = 1$, where the hook deletes the component along $\hat{v}_{\text{dim}}$ exactly (Eq. (2)). The effect is present at exact projection and grows with α: 11 of the 18 SED targets shift significantly in the predicted direction at $\alpha = 1$, and the strongest CAR cells already register, with Qwen3-8B's race target falling by 0.059 on race$\times$age and 0.019 on race$\times$gender.

Projection-magnitude-matched sham. The original random-direction reference matches the direction vector's norm, but the hook normalizes that vector before projection. We therefore rerun a stricter sham whose random-direction perturbation matches the real per-token magnitude $\alpha|h^{\top}\hat{v}|$. This control is generated end-to-end as its own run, so its ablation arm differs from Table 10 in the third decimal. At exact projection the matched sham moves SED-description by $+0.015$ against the ablation's $+0.079$ on Qwen3-8B race$\times$age, and the ablation's target-CAR reductions (-0.060 and -0.018 on the two Qwen3-8B race cells) do not appear under it ($+0.009$ and $+0.005$). At $\alpha = 5$ the sham produces nonzero semantic drift, while the Qwen3-8B target-CAR reductions remain specific to the learned direction. The large-strength semantic effect should therefore be read as a mixture of direction-specific removal and generic perturbation, while the exact-projection result supplies the cleaner causal test.

H MMLU Evaluation and Vector-Source Robustness

This appendix documents two supporting details for the direction ablation of §4.3: the protocol behind the MMLU capability-preservation check and a robustness comparison showing that the ablation's effect does not depend on the specific vector source used to define the ablated direction.

H.1 MMLU protocol

The capability check evaluates each of the three ablation models (Llama-3-8B, Mistral-7B, Qwen3-8B) on five MMLU subjects—elementary mathematics, college biology, high-school world history, philosophy, and sociology—at 100 questions each, for 500 questions per condition, posed in the standard few-shot MMLU format; accuracy is the macro-average over the five per-subject accuracies. We evaluate no intervention (*baseline*) and the direction ablation at $\alpha \in \{1, 3, 5\}$. Baseline macro-accuracy is 0.704 on Llama-3-8B, 0.616 on Mistral-7B and 0.634 on Qwen3-8B. At $\alpha = 1$ the largest change is -0.006 (Qwen3-8B); at $\alpha = 5$ it is -0.028, again on Qwen3-8B, while Llama and Mistral stay within 0.006 of baseline at every strength. For every ablation condition the intervention is *active during answer generation*—the same forward pre-hook, at the same $L^{\star}$ hook layers, that produces the behavioral effect—so the check measures the intervention's true collateral cost on general reasoning. Llama-3-8B is evaluated with the few-shot block as a plain continuation rather than wrapped in its chat template, whose turn structure breaks the benchmark's exemplars-then-question

Model	Pair	target	$\alpha^\star$	$\Delta\text{CAR}_{(t)}$	95% CI	Prompt Δ
llama-3-8b	race × gender	race	2.0	-0.030^{*}	$[-0.057, -0.006]$	$+0.025$
		gender	0.5	$+0.014$	$[+0.006, +0.022]$	-0.076
	race × age	race	0.5	-0.009	$[-0.020, +0.003]$	-0.004
		age	0.5	$+0.018$	$[+0.007, +0.030]$	-0.073
	gender × age	gender	0.5	$+0.006$	$[-0.003, +0.015]$	-0.066
		age	0.5	$+0.002$	$[-0.012, +0.015]$	-0.054
mistral-7b	race × gender	race	1.0	$+0.006$	$[+0.001, +0.011]$	$+0.005$
		gender	5.0	$\mathbf{-0.169}^{***}$	$[-0.193, -0.146]$	$+0.022$
	race × age	race	0.5	$+0.001$	$[-0.003, +0.004]$	$+0.011$
		age	0.5	-0.002	$[-0.008, +0.003]$	-0.012
	gender × age	gender	5.0	$\mathbf{-0.039}^{***}$	$[-0.055, -0.022]$	$+0.053$
		age	5.0	-0.024^{***}	$[-0.038, -0.011]$	-0.024
qwen3-8b	race × gender	race	5.0	$\mathbf{-0.092}^{***}$	$[-0.109, -0.076]$	-0.044
		gender	1.0	-0.002	$[-0.012, +0.007]$	$+0.011$
	race × age	race	4.0	$\mathbf{-0.130}^{***}$	$[-0.140, -0.121]$	-0.049
		age	1.0	$+0.009$	$[-0.002, +0.019]$	$+0.040$
	gender × age	gender	5.0	-0.030^{**}	$[-0.048, -0.012]$	$+0.007$
		age	5.0	-0.013	$[-0.035, +0.008]$	$+0.019$

Table 9: Full CAR ablation grid at `vk=dim`, both ablation sides per pair (rows labeled by the ablated *target* dimension), each cell at its own $\alpha^\star$. $\Delta\text{CAR}_{(t)}$ is the change in the target dimension's taxonomy CAR after ablation (negative = stereotype-labeled items removed); stars are from the one-sided Wilcoxon test on the target dimension's per-sample ΔCAR ($^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$); brackets are bootstrap 95% CIs. "Prompt Δ" is the matched "ignore-demographic" baseline on the same target. Bold marks the largest significant CAR reductions.

format and collapses the baseline to near chance; under the plain format its baseline is 0.704.

The explicit "ignore-demographic" prompt of §4.3 is excluded from this comparison because its effect on Qwen3-8B is anomalous: it *raises* macro-accuracy from 0.634 to 0.734, nearly uniformly across all five subjects (+0.03 to +0.17). A content-irrelevant instruction about demographics has no mechanism to improve elementary mathematics and philosophy alike; the pattern is instead consistent with a prompt-format artifact, where adding a system prompt changes the chat-template structure that Qwen3's answer formatting is sensitive to, and the other two models move negligibly under the same prompt (Llama +0.002, Mistral −0.004). We therefore do not read the prompt condition as evidence of a capability change in either direction.

H.2 Vector-source robustness

Three ways to define the ablated direction. The ablation of §4.3 projects out the unit direction of the *pure single-dimension contrast* (`vk=dim`): for the ablated dimension A, $v_A = \mathbf{A}(C_{\text{pure-}A}) - \mathbf{A}(C_{\text{neu}})$ at $L^\star$, the dose-matched raw single-cue activation contrast used throughout the paper (§3.4). Two alternative estimators of "the A direction" are available. `vk=intersect` takes the mixed-versus-dose-matched-pure contrast ($\mathbf{A}(C_{\text{mixed}}) - \mathbf{A}(C_{\text{pure-}B})$), so it isolates A's contribution from *within* a two-cue context and thus carries additional intersectional signal. `vk=explicit` takes the contrast between an explicitly-stated-demographic context and the neutral context. All three name the same nominal direction by a different activation difference; §4.3 uses `dim` because it is the simplest such difference, is the most aligned with the activation-steering prior art, and—unlike intersect— requires no dependence on the mixed-context data.

The effect is qualitatively source-invariant (Table 11). Table 11 reruns the ablation under each of the three vector sources: three models × three pairs, ablate_A on description-SED, each source at its best sweep strength. The qualitative picture of §4.3 reproduces under all three: on the strong-personalization cells the target-dimension shift is positive and Wilcoxon-significant in the predicted direction under every source, and the underlying sweeps rise monotonically with α while the original random-direction reference stays near zero. The central claim that projecting out the cue-induced dimension direction suppresses the target-dimension response is therefore not an artifact of the particular contrast chosen to define that direction.

What differs across sources. The three sources are not identical in magnitude or in the relative ordering of the two dimensions within a pair, and

Model	Pair	target	$\Delta_{\mathrm{pure}(t)}$	d	$\Delta\mathrm{CAR}_{(t)}$
llama-3-8b	race $\times$ gender	race	$+0.018^{**}$	$+0.15$	-0.017
		gender	-0.002	-0.03	$+0.027$
	race $\times$ age	race	$+0.018^{***}$	$+0.17$	$+0.008$
		age	$+0.015^{***}$	$+0.15$	$+0.026$
	gender $\times$ age	gender	-0.002	-0.03	$+0.006$
		age	$+0.010^{*}$	$+0.11$	$+0.016$
mistral-7b	race $\times$ gender	race	$+0.025^{***}$	$+0.24$	$+0.006$
		gender	$+0.028^{***}$	$+0.31$	-0.009
	race $\times$ age	race	$+0.026^{***}$	$+0.26$	$+0.001$
		age	$+0.007^{*}$	$+0.08$	-0.002
	gender $\times$ age	gender	$+0.000$	$+0.01$	-0.005
		age	-0.011	-0.15	-0.011^{**}
qwen3-8b	race $\times$ gender	race	$+0.005$	$+0.04$	-0.019^{***}
		gender	$+0.010^{*}$	$+0.10$	-0.002
	race $\times$ age	race	$+0.079^{***}$	$+0.33$	-0.059^{***}
		age	$+0.022^{***}$	$+0.15$	$+0.009$
	gender $\times$ age	gender	$+0.008$	$+0.07$	$+0.018$
		age	$+0.011$	$+0.07$	$+0.003$

Table 10: All 18 ablation cells at exact projection ($\alpha = 1$), vk=`dim`: the SED-description shift away from the target reference $\Delta_{\mathrm{pure}(t)}$ with Cohen's d, and the target-taxonomy $\Delta\mathrm{CAR}_{(t)}$. Stars: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$, from the same tests as Tables 2 and 9 (SED: one-sided t, predicted positive direction; CAR: one-sided Wilcoxon, predicted negative direction); stars are shown only for movements in the predicted direction.

Model	Pair	`dim`	`intersect`	`explicit`
llama-3-8b	race×gender	$+0.086^{***}$	$+0.078^{***}$	$+0.056^{***}$
	race×age	$+0.078^{***}$	$+0.062^{***}$	$+0.041^{***}$
	gender×age	$+0.003$	$+0.006$	$+0.016^{**}$
mistral-7b	race×gender	$+0.226^{***}$	$+0.322^{***}$	$+0.152^{***}$
	race×age	$+0.154^{***}$	$+0.375^{***}$	$+0.022^{***}$
	gender×age	$+0.028^{***}$	$+0.025^{***}$	$+0.004^{*}$
qwen3-8b	race×gender	$+0.176^{***}$	$+0.103^{***}$	$+0.118^{***}$
	race×age	$+0.306^{***}$	$+0.113^{***}$	$+0.053^{***}$
	gender×age	$+0.078^{***}$	$+0.150^{***}$	$+0.027^{***}$

Table 11: Vector-source robustness: the ablate_A target-dimension shift on description-SED at each source's best sweep strength, per (model, pair). Stars: one-sided Wilcoxon in the predicted direction, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. The `dim` column matches the corresponding rows of Tables 2 and 8.

we do not claim they are. `intersect` produces the largest race effects on Mistral, where it drives the race target substantially higher than `dim` on both race×gender and race×age, consistent with its carrying extra mixed-context signal, though on the other two models `dim` is the stronger race-target source; `explicit` is the most variable across cells and is the weakest on some, falling to near the sham for the race target on Mistral race×age. It can also move a target the primary source does not: on Qwen3-8B race×age the age target's CAR falls by 0.155 at $\alpha = 5$ under `explicit` but shows no reduction under `dim`, the clearest single case of source dependence. The ceiling-of-sweep behavior ($\alpha^{\star}$ at or near 5) is shared by all three; the stricter matched-perturbation sham is analyzed in Appendix G. We adopt `dim` as the primary source not because it is uniformly the strongest—it is not—but because it is the simplest and most prior-art-aligned estimator and introduces no methodological dependence on the intersectional data; the two alternatives serve to corroborate, rather than to change, the conclusion of §4.3.

The prompt can backfire where the ablation does not. Figure 6 isolates the clearest case from the vk=`dim` grid, in SED-description terms: on Mistral race×gender both ablation targets climb monotonically with α, while the two explicit-prompt baselines sit flat near zero. The race-side prompt is slightly negative (-0.008), so telling the model to ignore race moves its recommendations *toward* the stereotype; the gender-side prompt is only marginally positive ($+0.014$). The sham references

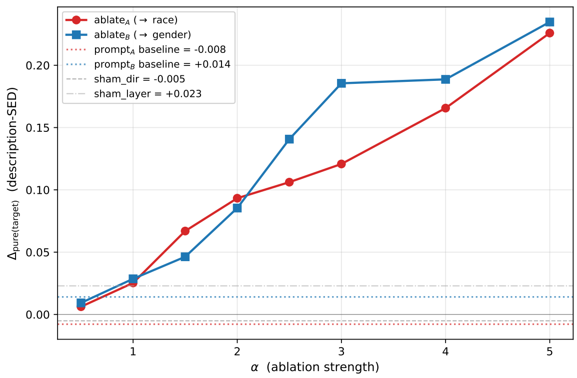

Figure 6: Direction ablation vs. prompt vs. sham on Mistral-7B race×gender, SED-description. Both ablation targets (ablate$_A$ →race, ablate$_B$ →gender) rise monotonically with α; the two "ignore-demographic" prompt baselines are flat and near zero, the race-side one slightly negative (that prompt moves the output toward the cue), and the original direction- and layer-shams hug zero.

hug zero. This is the SED-description view of the prompt comparison summarized in §4.3, where the main text reports the CAR view (Fig. 4).

Output-collapse diagnostic. To test whether reduced personalization merely reflects malformed or collapsed recommendations, we examine list length, within-response title repetition, corpus-level diversity, and overlap with the unablated response. At exact projection ($\alpha = 1$), responses retain five-item lists with within-response distinct-title ratios of 0.999–1.000; their corpus-level diversity is at least as high as under the prompt in every model. They overlap with the unablated recommendations by 0.76–0.89, at least as much as under the prompt baseline (0.69–0.84). At $\alpha = 5$, overlap falls to 0.39–0.59, indicating that amplified removal causes broader output changes. These diagnostics rule out gross formatting or repetition collapse at exact projection but do not constitute a user-level evaluation of recommendation quality.

I Selectivity of the Direction Ablation

§4.3 reports whether removing one dimension's direction spares the co-present dimension. This appendix records why that question can only be answered on the taxonomy metric, and how the answer varies by model.

Semantic distance cannot express selectivity. For an ablate$_A$ intervention, the natural semantic selectivity measure is $\Delta_{\text{pure}A} - \Delta_{\text{pure}B}$, the shift away from the A-target minus the shift away from the B-target. This quantity is uninformative by construction: any edit that changes the response at all moves it away from both pure references, so the two shifts rise together whatever the edit removed. Empirically they do, at every α we sweep, including exact projection. Semantic distance therefore registers that the intervention acted, not which dimension it acted on. CAR counts content per dimension and can separate the two.

Confinement holds on two models of three. On Mistral-7B, ablating the gender direction lowers gender-tagged content by 0.169 on race×gender and 0.039 on gender×age while race- and age-tagged content moves by $+0.004$ and -0.003; across the full sweep of both cells the partner dimension's tagged rate never falls at any α. Llama-3-8B shows the same in the other direction: ablating race lowers race-tagged content by 0.030 with gender-tagged content at $+0.002$, again never falling across the sweep. Neither operating point was chosen for selectivity—each is the α that maximizes the target reduction—so these are unselected demonstrations that the removal can be confined to one dimension.

Qwen3-8B is the exception. At its taxonomy-optimal strengths the two dimensions fall together: ablating race on race×age lowers race-tagged content by 0.130 and age-tagged content by 0.104, and on gender×age the off-target move (-0.128, $p = 2.9 \times 10^{-15}$) exceeds the target one (-0.030). On this model the intervention is better described as suppressing personalization along a shared identity axis than as a dimension-specific edit. Target and partner CAR values for all three models are released with the code.

Why confinement is partial. That the removal is dimension-specific on some models and not others is consistent with the composition result of §4.2. The mixed-cue direction is substantially but not fully a linear combination of the single-cue directions ($\bar{\rho} \in [0.576, 0.704]$), so the per-dimension directions are not mutually orthogonal, and a rank-one projection along one of them removes some of the other. We read this as cue-associated directions inhabiting overlapping subspaces rather than clean orthogonal axes, consistent with the linear-representation and superposition literature (Arditi et al., 2024; Marshall et al., 2024). Making the confinement hold across models remains open.